



THE THINGS
NETWORK



DATACAKE

I

Erste Schritte im IoT mit LoRaWAN

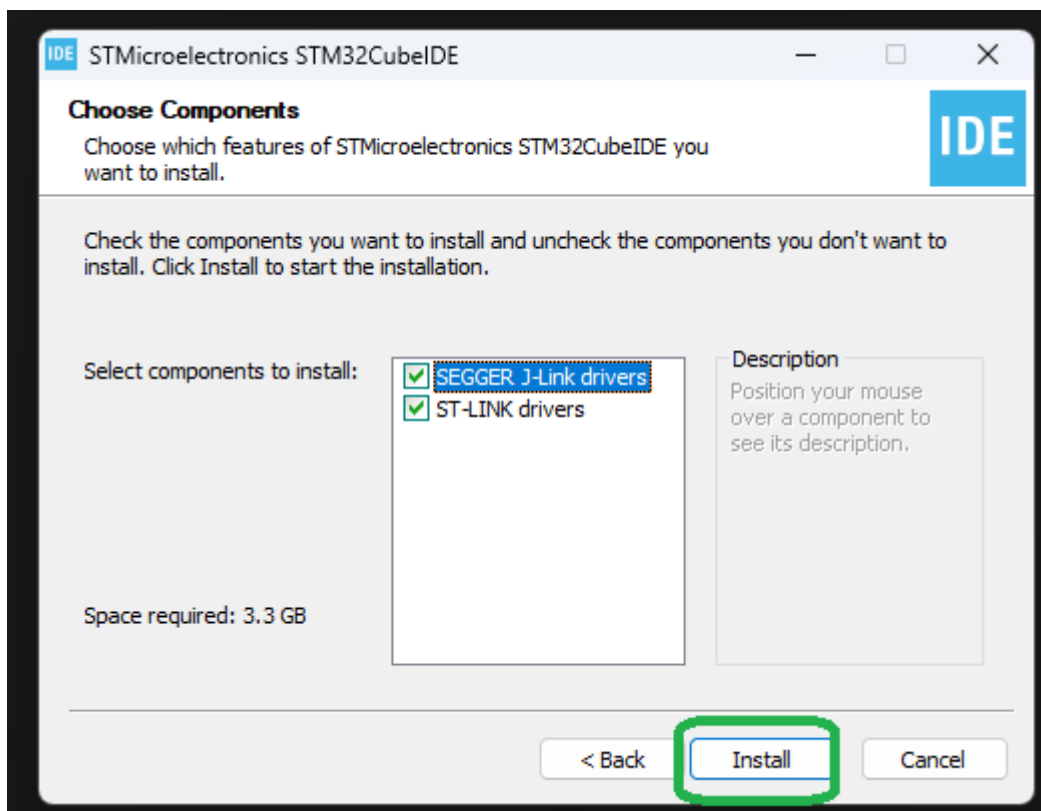
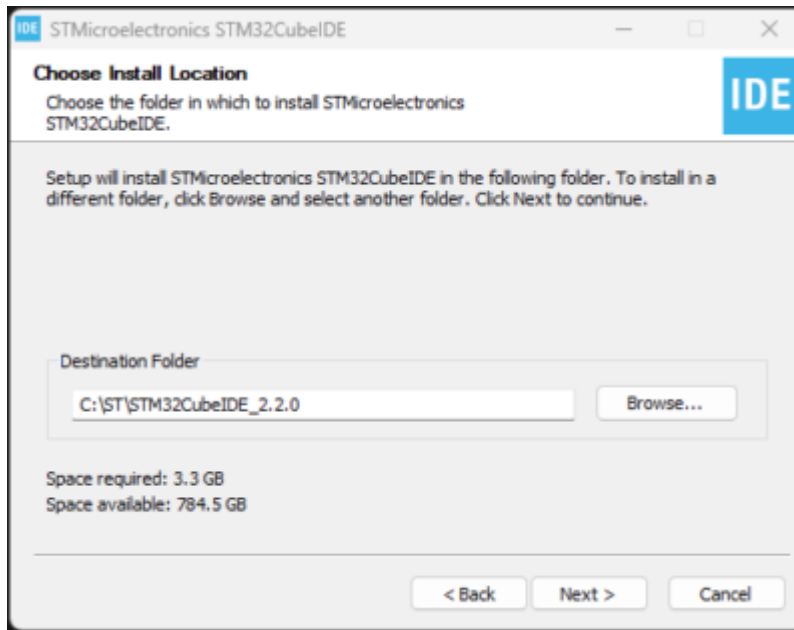
Entwicklungsumgebung

- Hardware/Firmware
 - Gateway(optional): [NUCLEO-F746ZG](#) + [LRWAN GS HF1 LoRa](#)
 - Entwicklungsboard: [STM32WL55JC1](#) : STM32WL55JC1x Firmware: V1.3.1 + [Treiber](#)
 - Apiar-IOT-PCB: Demnächst verfügbar.
- *Software:*
 - [STMCubeIDE](#) (V:2.2.0)
 - [STM32CubeMX](#) (V:6.18.1)
 - [STMCubeWL](#) (V1.6.0/1: Bibliothek für STM32WL55JC)
 - Entwicklungsboard STM32WL55JC1x Firmware: V1.3.

Vorbereitung der Umgebung (Toolchain)

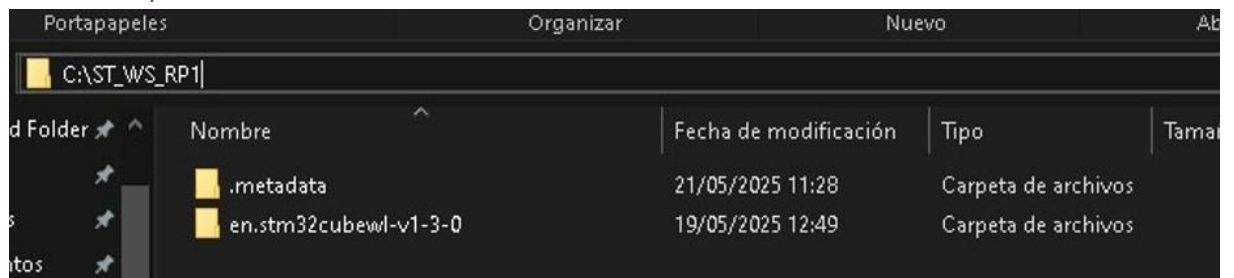
Es wird empfohlen, denselben Ordnerpfad beizubehalten und jeden Schritt genau zu befolgen.

- Lade die integrierte Entwicklungsumgebung (IDE) "[STM32CubeIDE](#)" herunter und installiere sie.

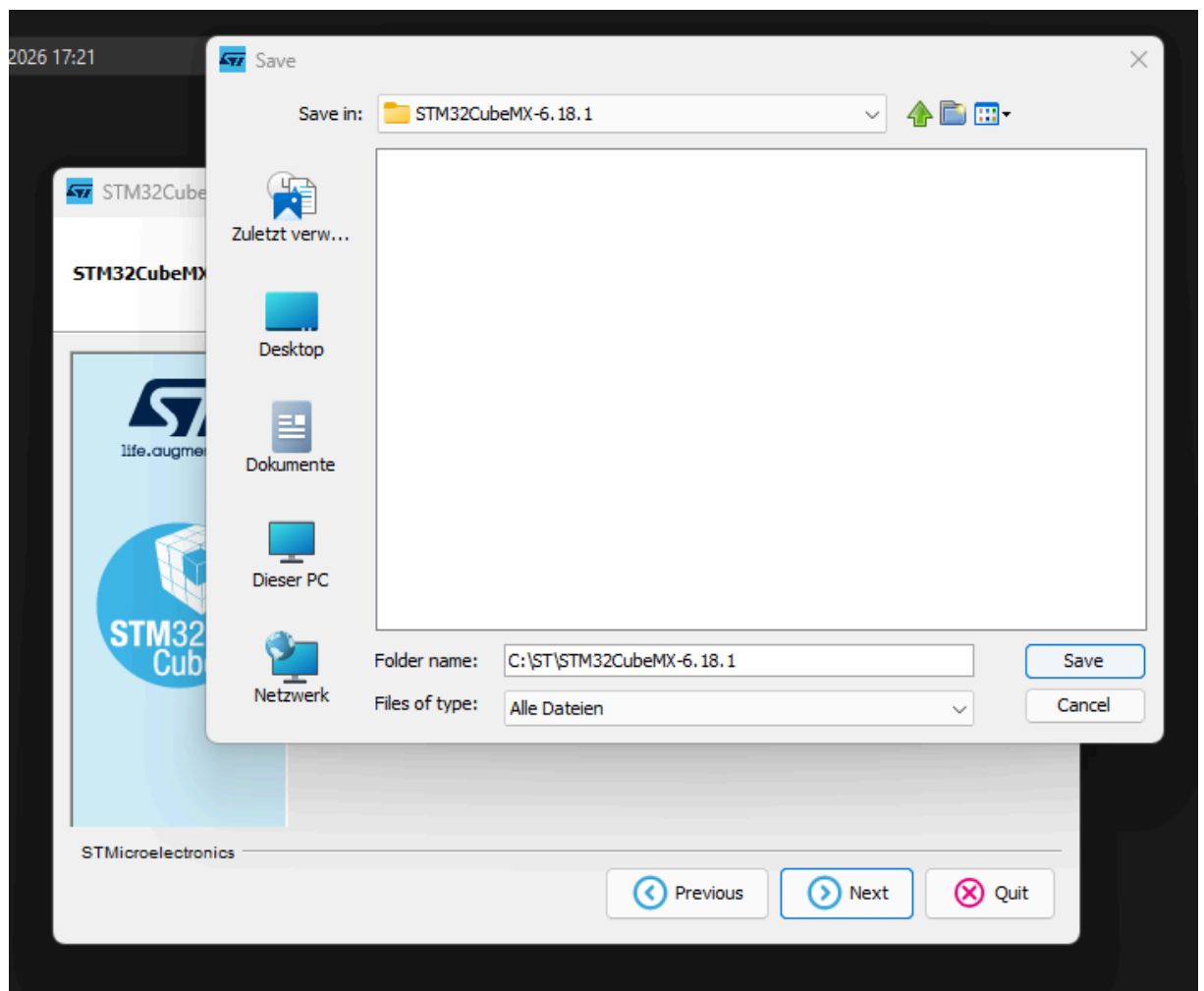


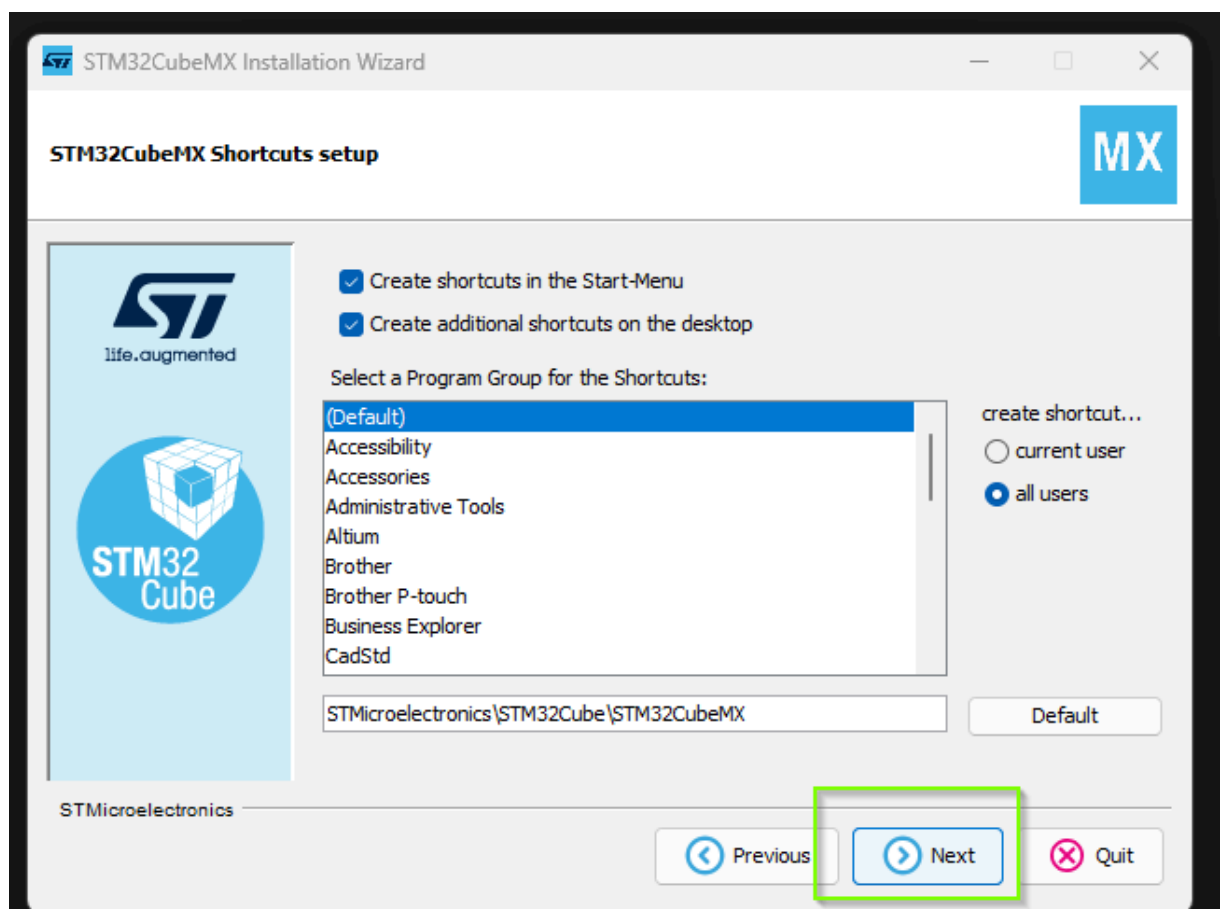
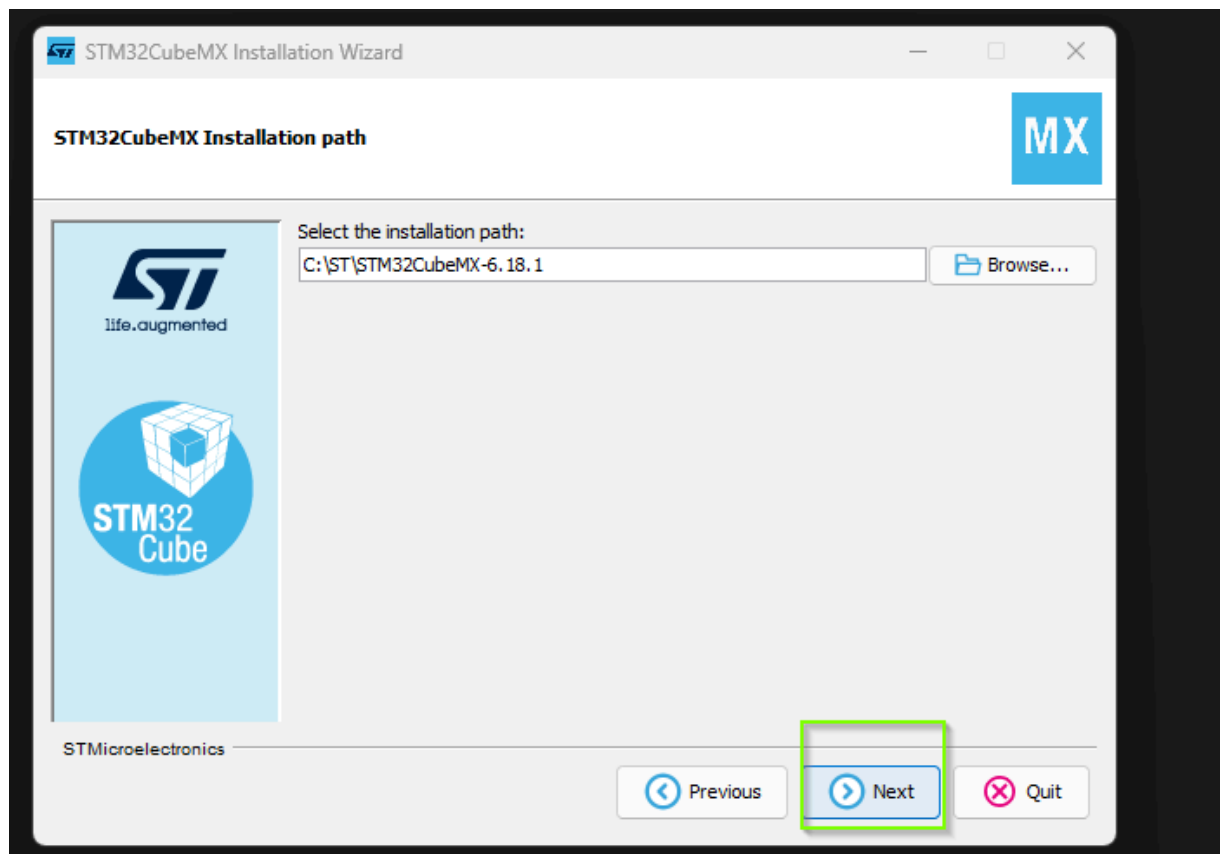
- Erstelle den Ordner "C:ST_WS_RP1".

Es wird empfohlen, denselben Ordner und denselben Pfad zu verwenden.

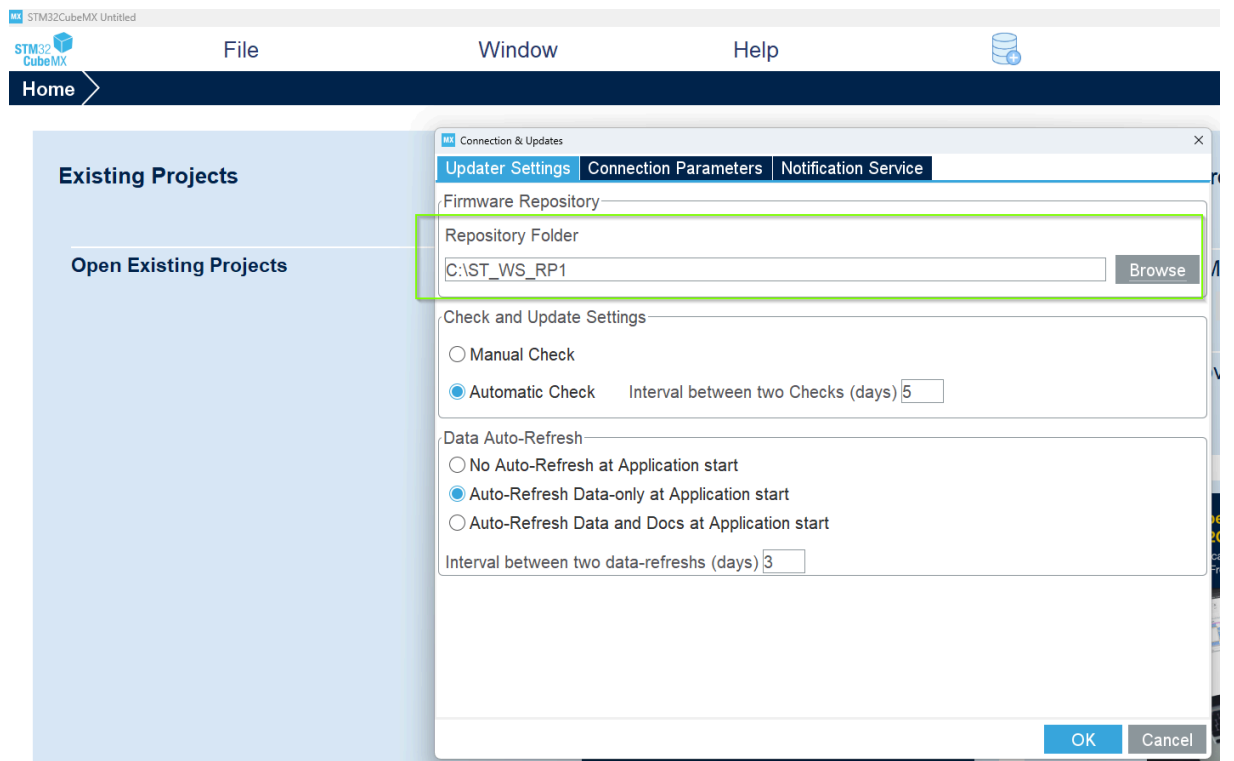
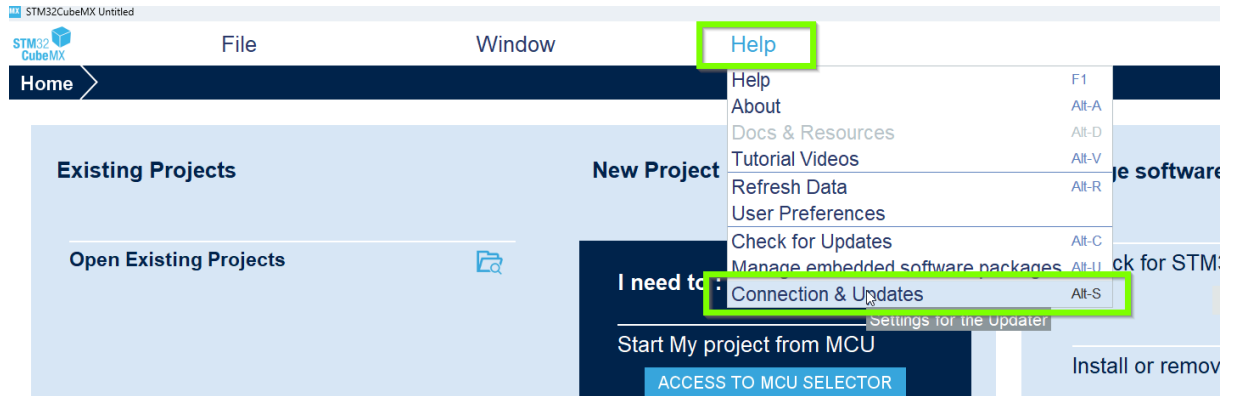


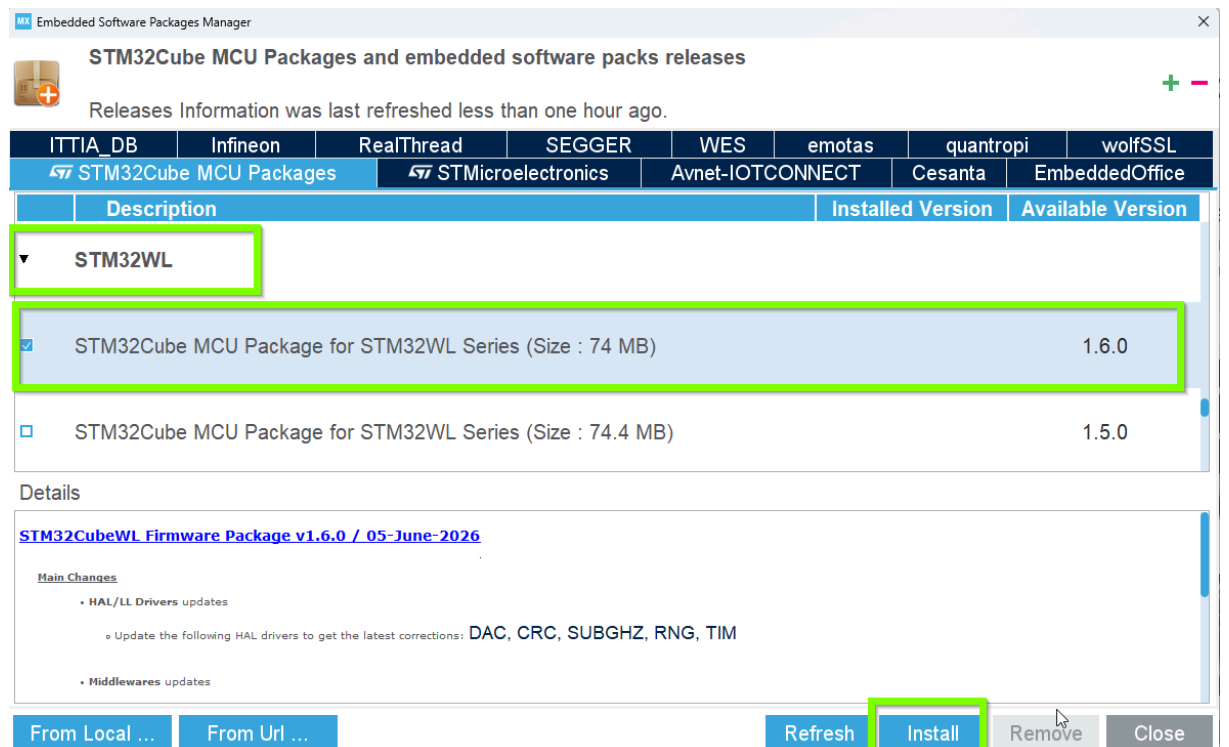
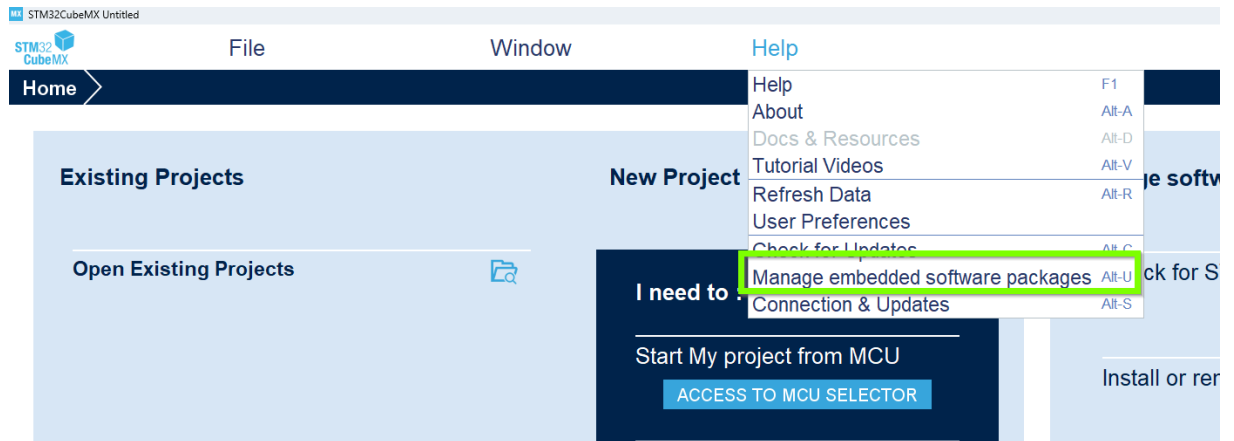
- Erstelle den Ordner "C:STSTM32CubeMX-6.18.1 ", lade [STM32CubeMX](#) (V:6.18.1) herunter und installiere es.





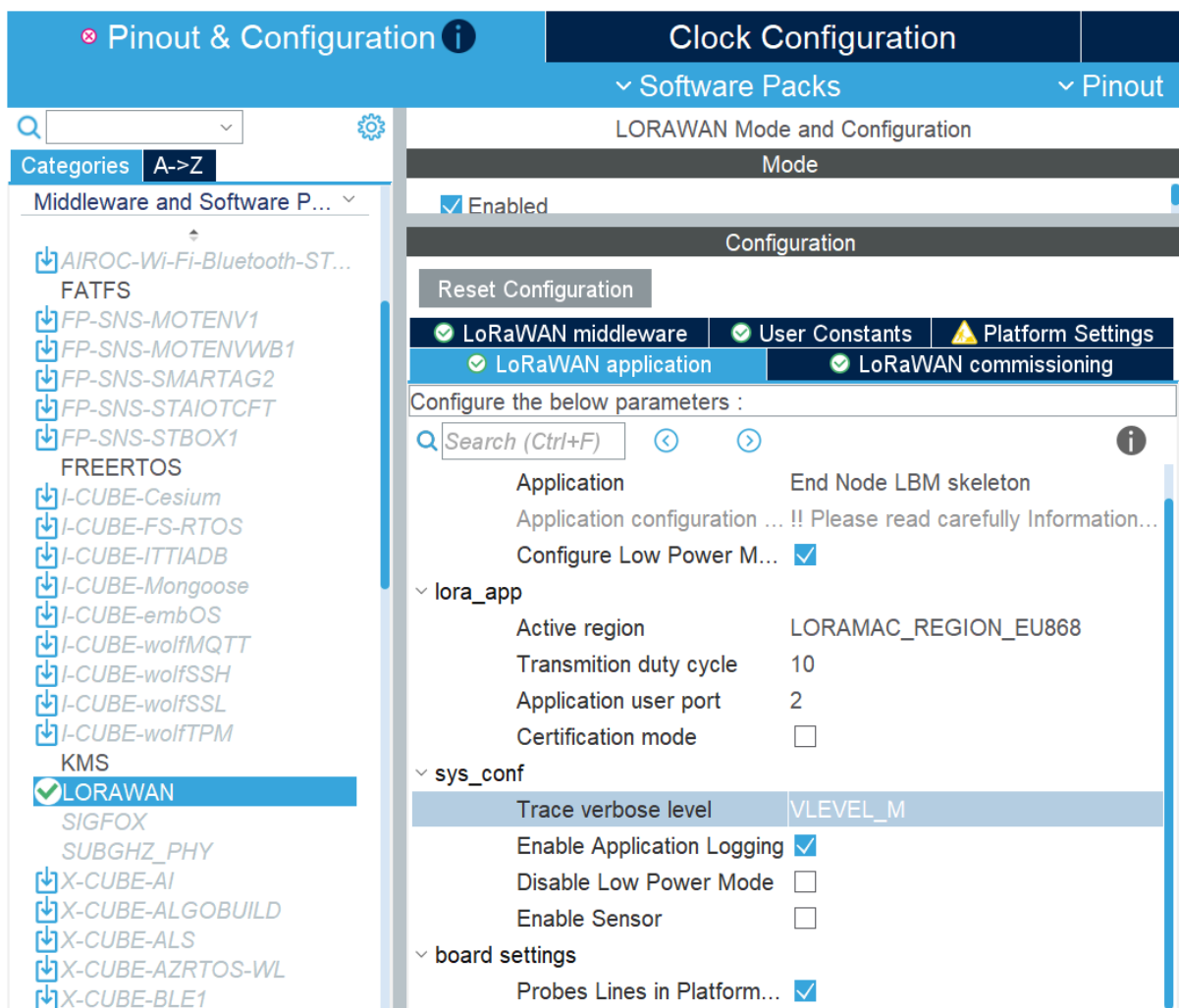
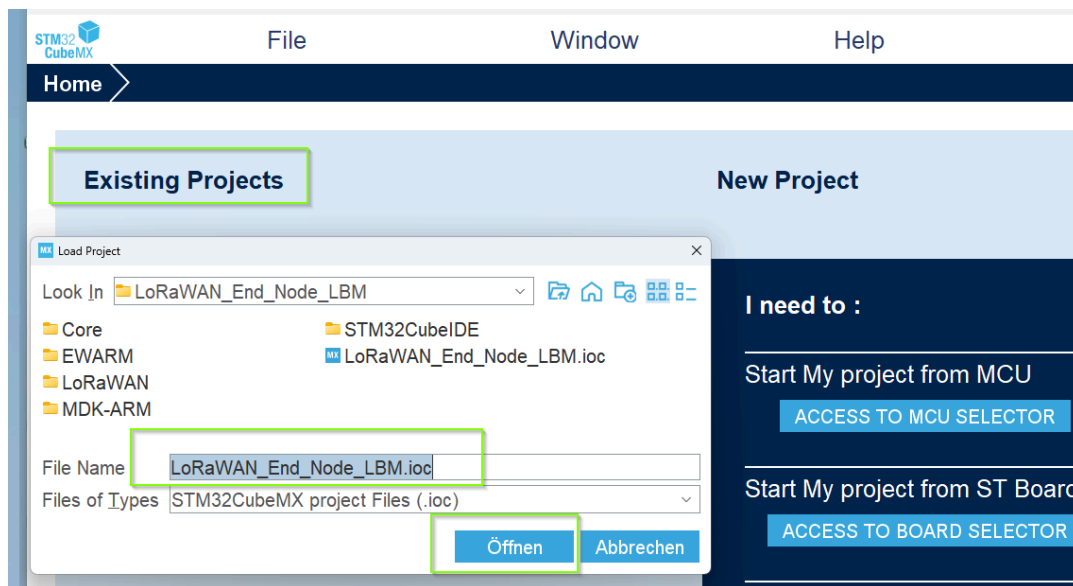
- Starte STM32CubeMX





LoRaWAN Konfiguration:

- Starte:
LoRaWAN_End_Node_LBM.iocC:ST_WS_RP1STM32Cube_FW_WL_V1.6.0Projects
NUCLEO-WL55JCAApplicationsLoRaWANLoRaWAN_End_Node_LBMLoRaWAN_End
Node_LBM.ioc



LoRaWAN commissioning:

- *App/Join EUI*: Ist auf dem Board zu finden.



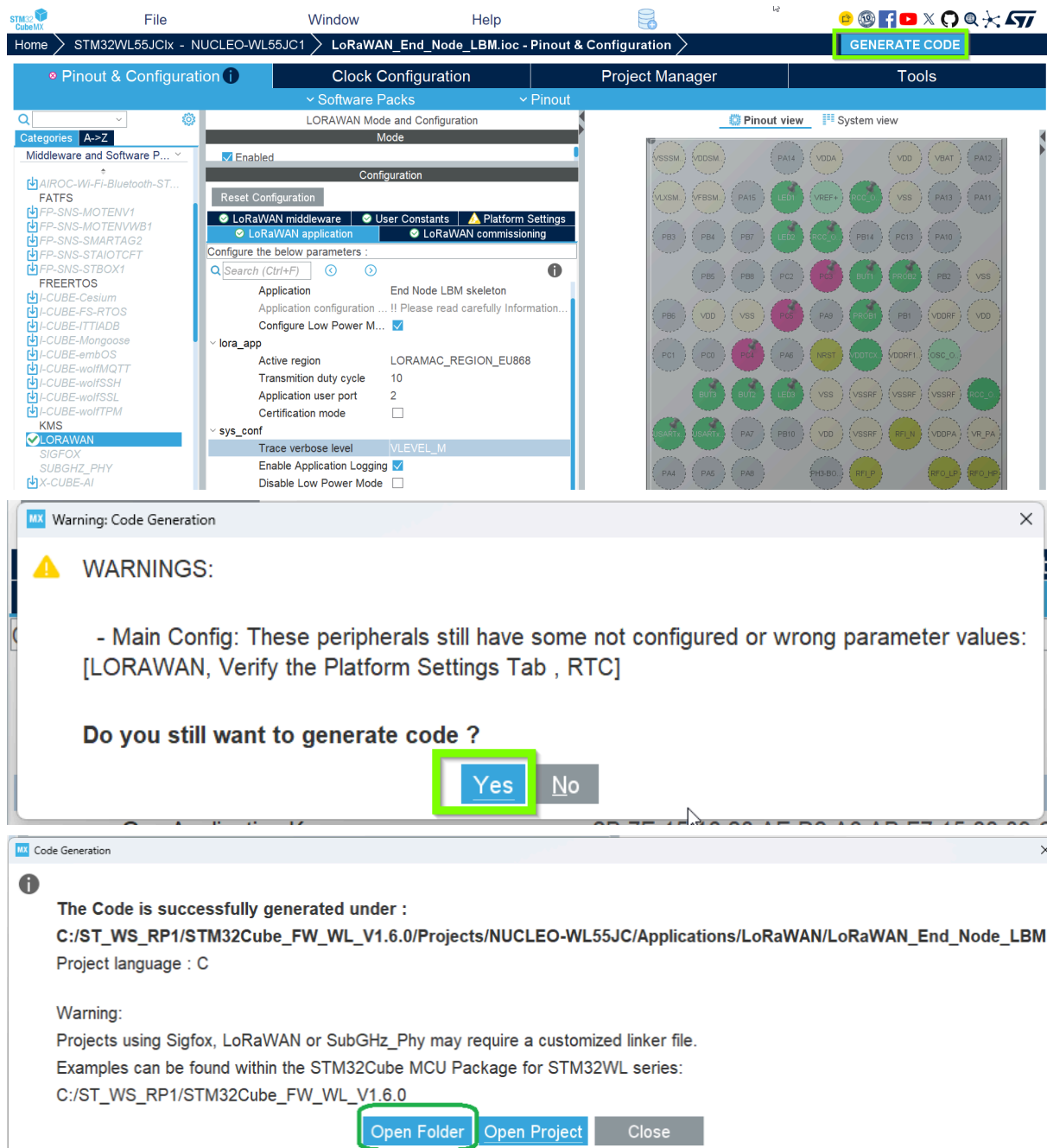
Wichtig: Das Feld "LORAWAN_device EUI" sollte NICHT als "00.00.00.00.00.00.00.00" definiert bleiben. Selbst wählen.

```
/*
 * End-device IEEE EUI (big endian)
 * When set to 00,00,00,00,00,00,00,00 DevEui is automatically set
 * with a value provided by MCU platform
 */
#define LORAWAN_DEVICE_EUI
00,00,00,00,00,00,00,00
```

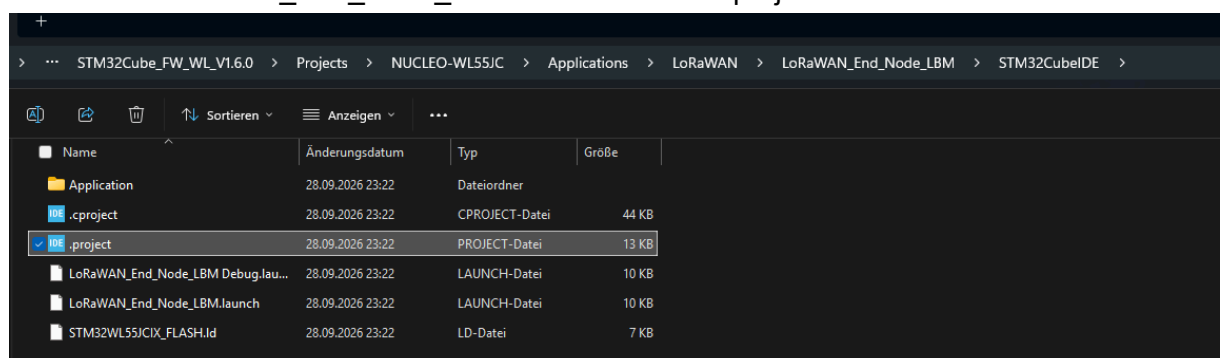
- “Application key” and “Network key”: selbst wählen

The screenshot shows the STM32CubeMX IDE interface for configuring a LoRaWAN project. The 'Pinout & Configuration' tab is active, displaying the 'LORAWAN Mode and Configuration' settings. The 'App/Join EUI' field is highlighted in green, showing the value '00,80,E1,15,05,00,D7,53'. The 'File' menu is open, and the 'Save Project' option is highlighted in green. The 'Save Project' option is also highlighted in green.

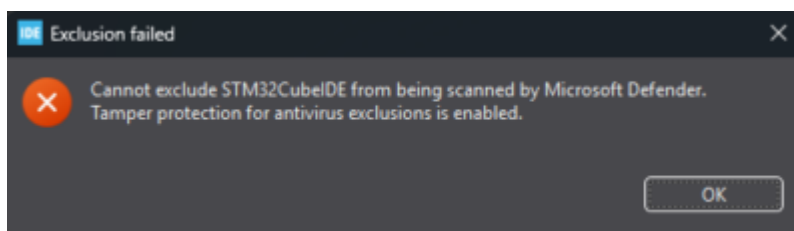
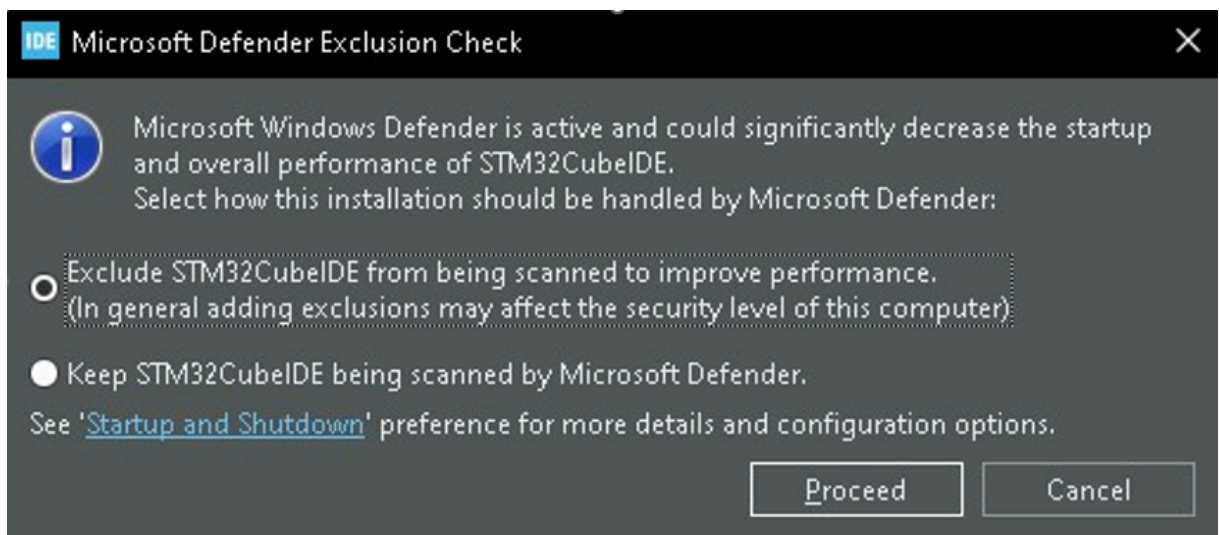
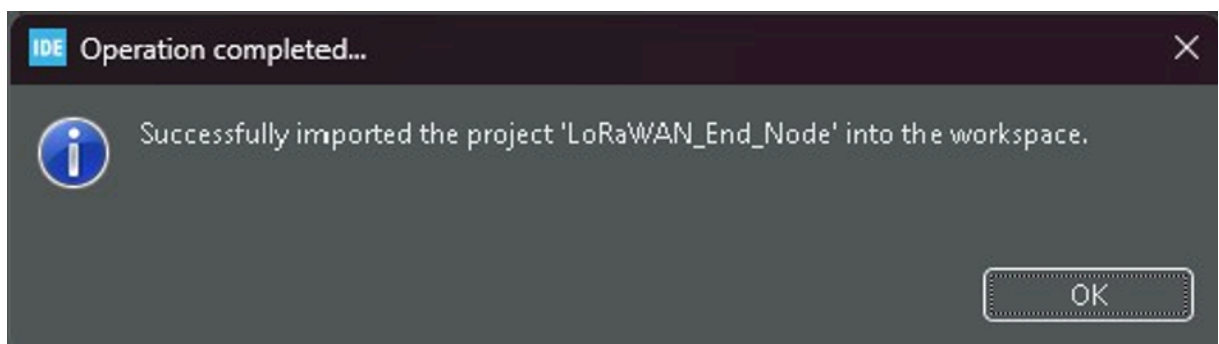
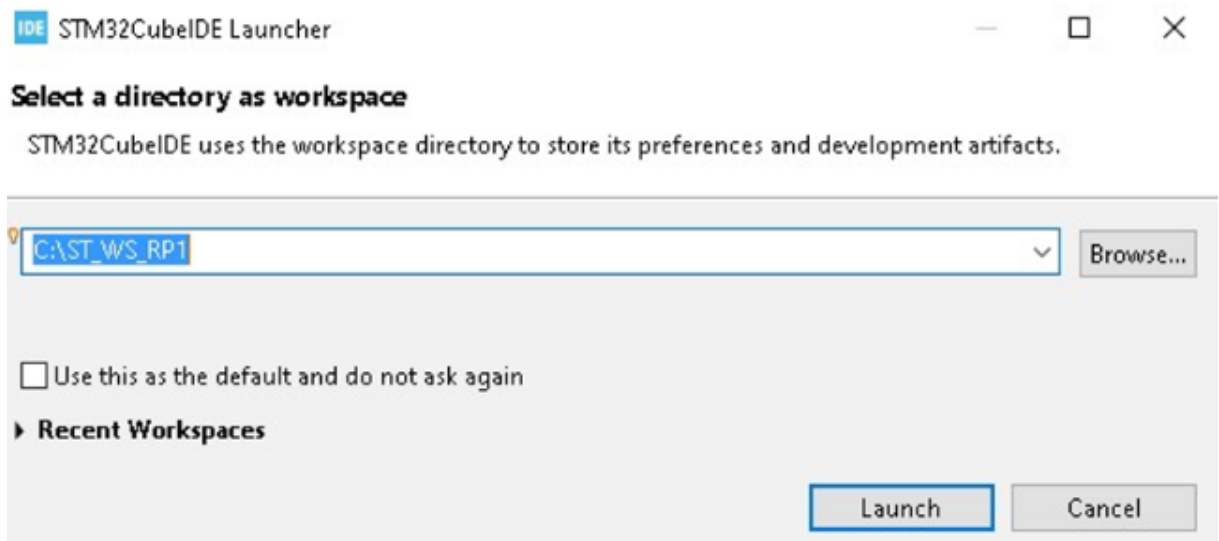
Parameter	Value
LoRaWAN device EUI	00,00,00,00,00,00,00,00
App/Join EUI	00,80,E1,15,05,00,D7,53
Gen Application Key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Device Address	00,00,00,00
Network session key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C
Application session key	2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,09,CF,4F,3C



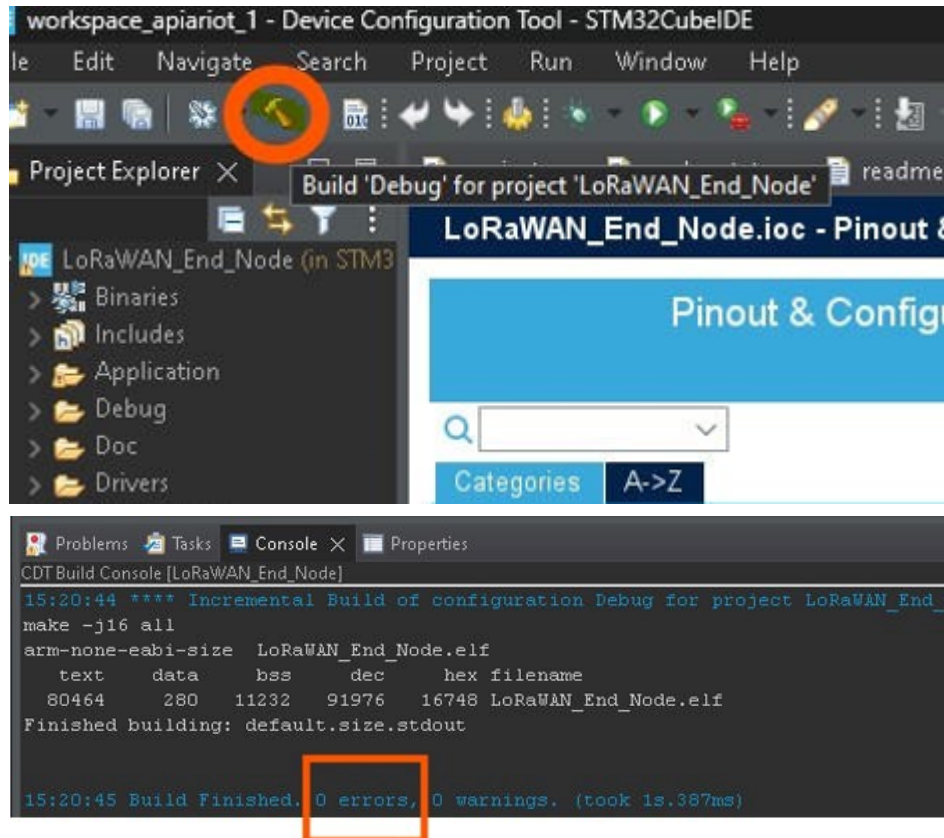
- Starte das Projekt mit STM32cubeIDE
"C:/ST_WS_RP1/STM32Cube_FW_WL_V1.6.0/Projects/NUCLEO-WL55JC/Applications/LoRaWAN/LoRaWAN_End_Node_LBM/STM32CubeIDE.project"



- Wähle als Workspace C:ST_WS_RP1



- Kompiliere das Projekt fehlerfrei :



Programmiercode-Änderung

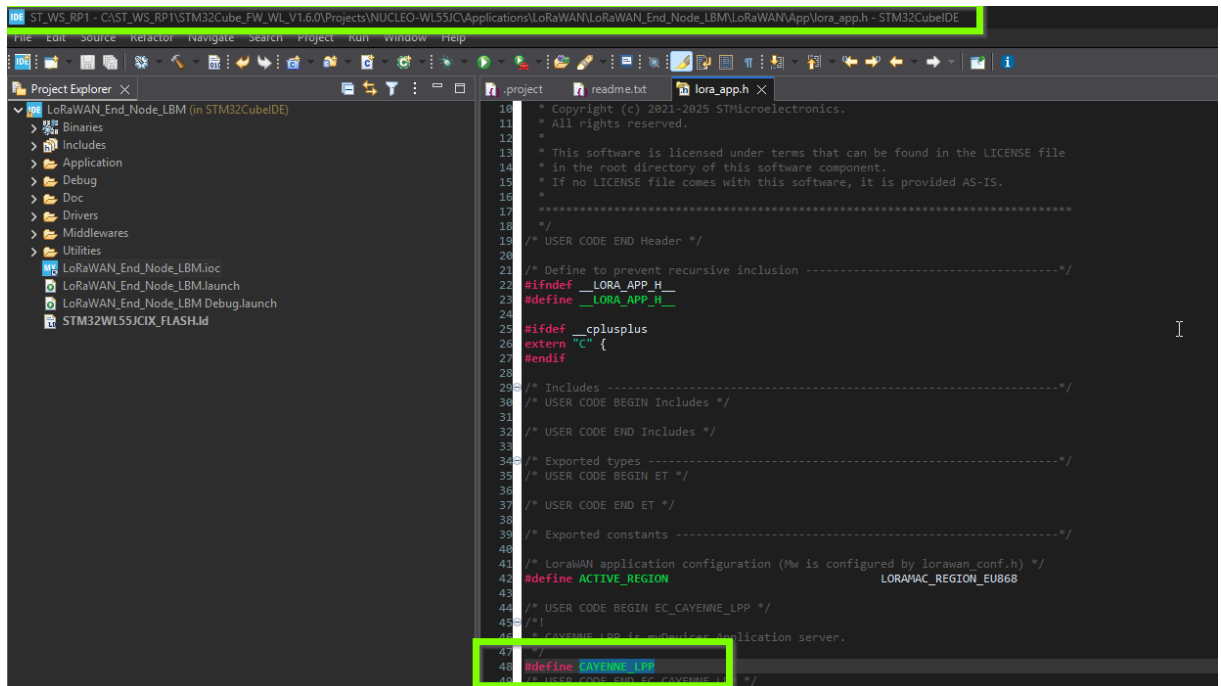
- Aktiviere "Low Power Payload" von Cayenne:
- Entkommentiere `/*#define CAYENNE_LPP` in `lora_app.h`*/:

#define CAYENNE_LPP

"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWANApp\lora_app.h"

Hinweis: Cayenne LPP ist ein kompaktes Datenformat zur Kodierung von Sensordaten für LoRaWAN. Mehr Infos unter diesem Link:

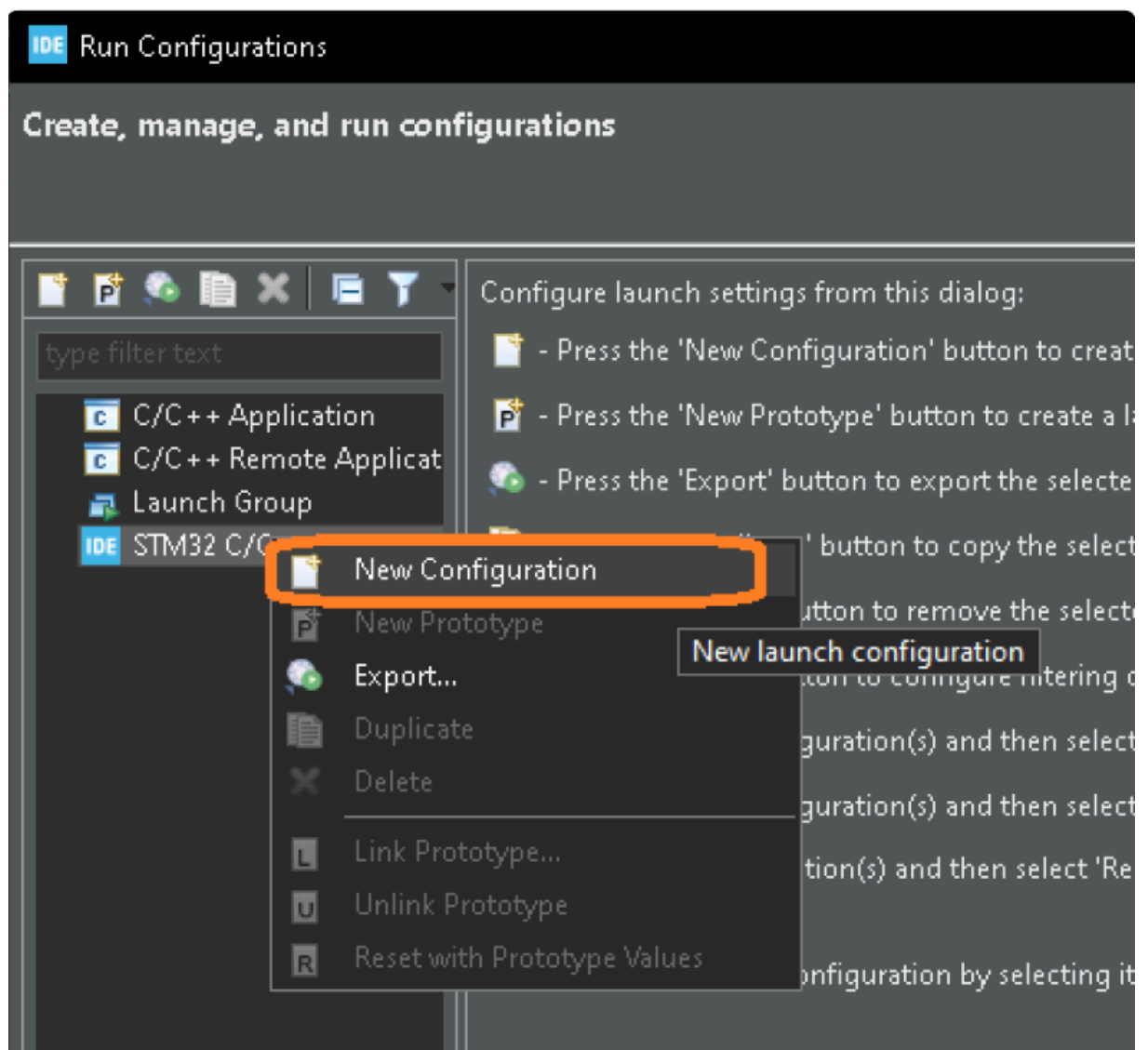
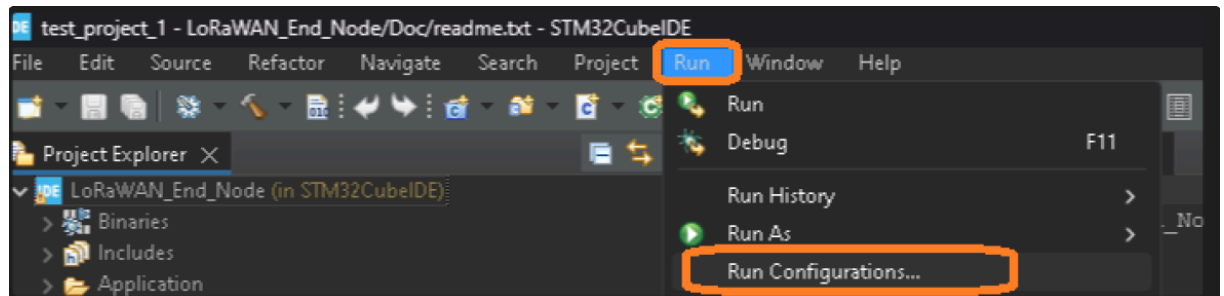
[Was ist "Cayenne Low Power Payload"? 🤔👉](#)



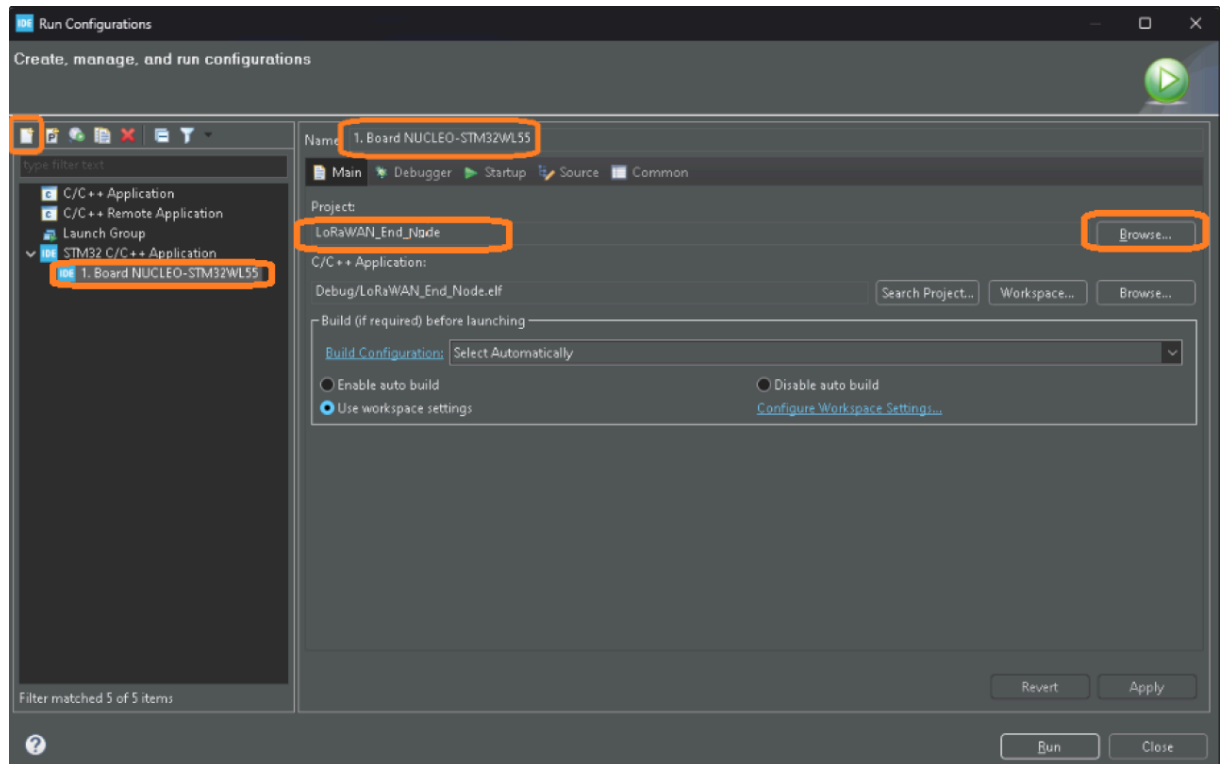
```
10  * Copyright (c) 2021-2025 STMicroelectronics.
11  * All rights reserved.
12  *
13  * This software is licensed under terms that can be found in the LICENSE file
14  * in the root directory of this software component.
15  * If no LICENSE file comes with this software, it is provided AS-IS.
16  *
17  * =====
18  */
19  /* USER CODE END Header */
20
21  /* Define to prevent recursive inclusion -----*/
22  #ifndef LORA_APP_H
23  #define LORA_APP_H
24
25  #ifdef __cplusplus
26  extern "C" {
27  #endif
28
29  /* Includes -----*/
30  /* USER CODE BEGIN Includes */
31
32  /* USER CODE END Includes */
33
34  /* Exported types -----*/
35  /* USER CODE BEGIN ET */
36
37  /* USER CODE END ET */
38
39  /* Exported constants -----*/
40
41  /* LoRaWAN application configuration (Mw is configured by lorawan_conf.h) */
42  #define ACTIVE_REGION LORAMAC_REGION_EU868
43
44  /* USER CODE BEGIN EC_CAYENNE_LPP */
45  /*
46  * CAYENNE_LPP is a Cayenne application server.
47  */
48  #define CAYENNE_LPP
49  /* USER CODE END EC_CAYENNE_LPP */
```

Board Einstellungen

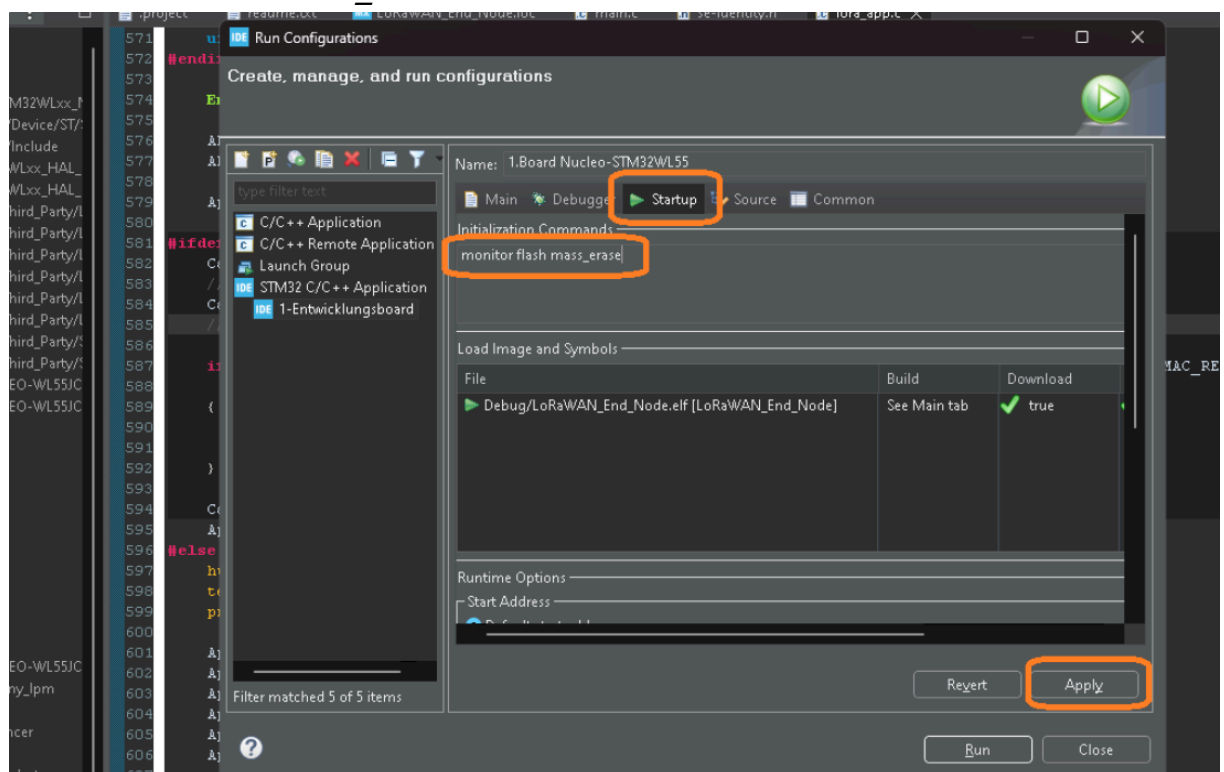
- Erstelle für das Entwicklungsboard "new konfiguration"



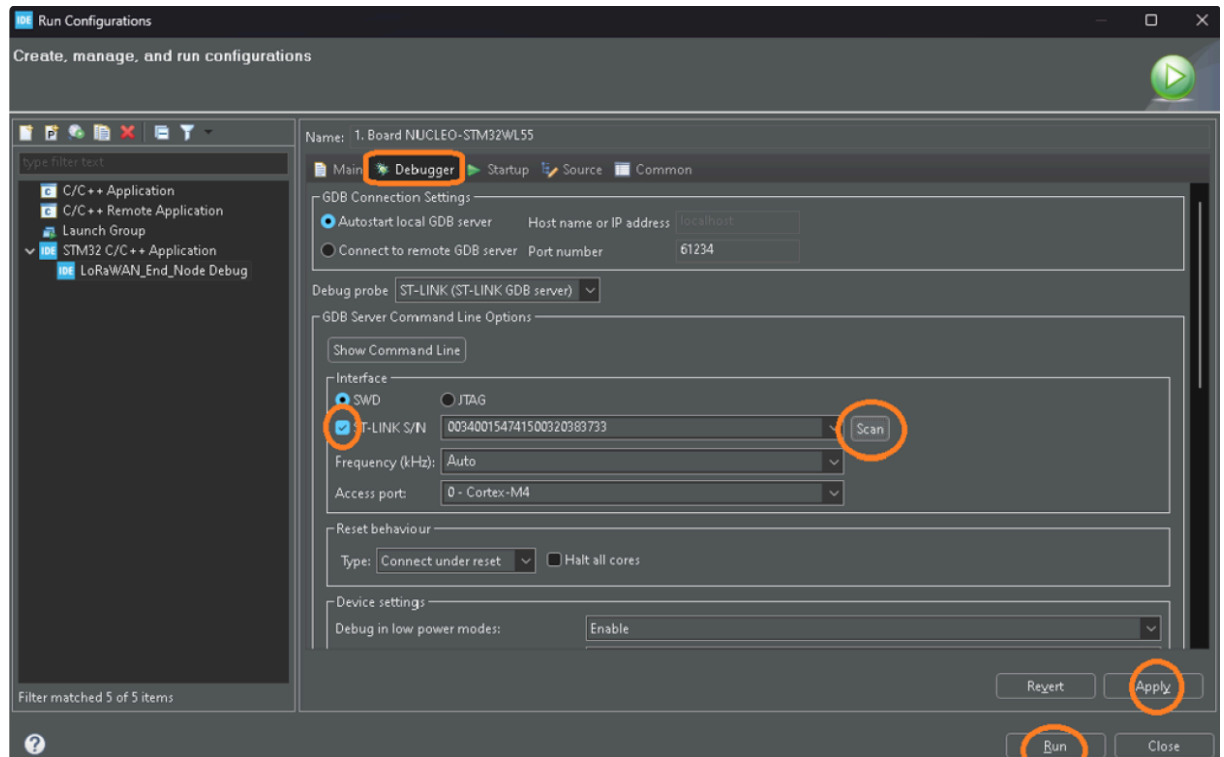
- Benenne dein Board NUCLEO-STIM32WL55



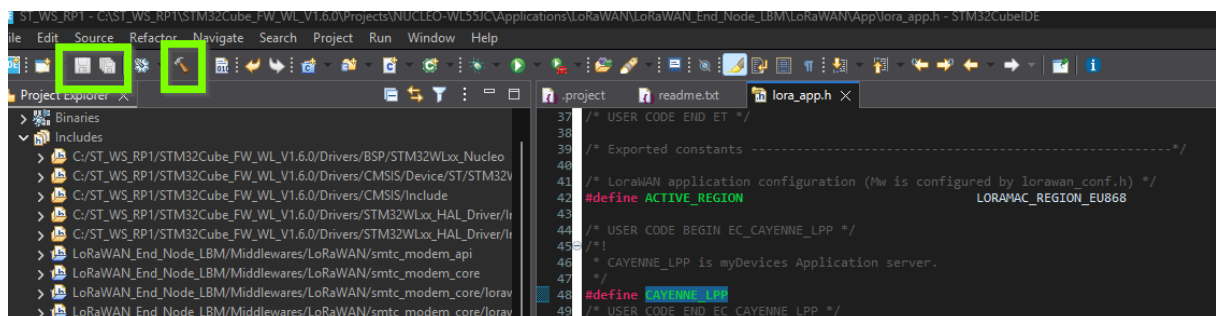
- Füge unter *Initialization Commands* folgenden Befehl hinzu:
"monitor flash mass_erase"



- Entwicklungsboard per USB mit dem PC verbinden

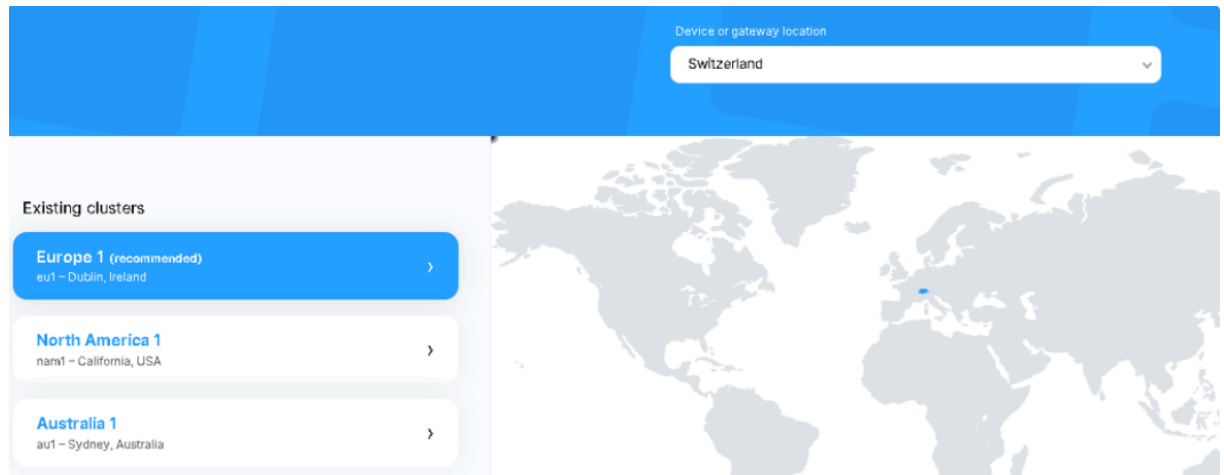


- Speichere die Konfiguration



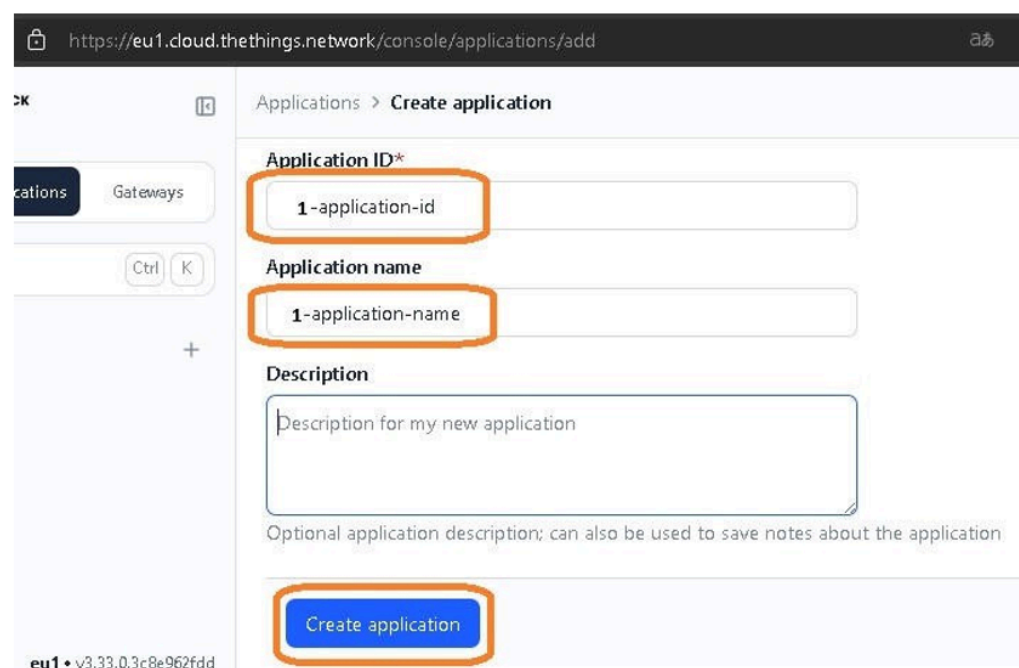
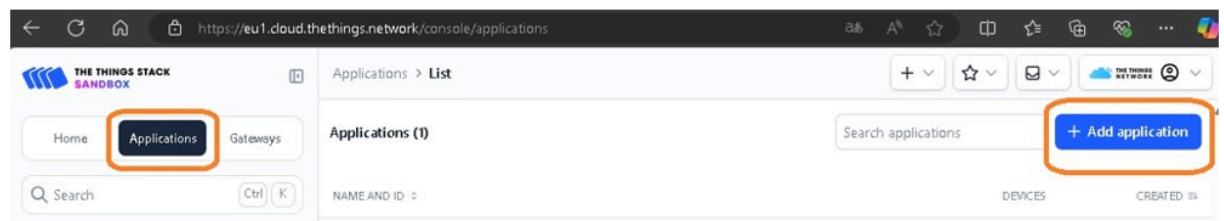
The Thing Network (TTN) / Stacks

- Erstelle ein Konto, um Zugriff auf , The things network zu erhalten.
- Wähle Cluster Europe 1 (Für Schweiz)



Cluster Für die Schweiz: Europe 1

- Erstelle eine Applikation



Create application

Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.

Learn more in our guide on [Adding Applications](#).

Application ID*

my-application-1-test

Application name

my application 1 test

Description

Description for my new application

Optional application description; can also be used to save notes about the application

Application properties

Labels

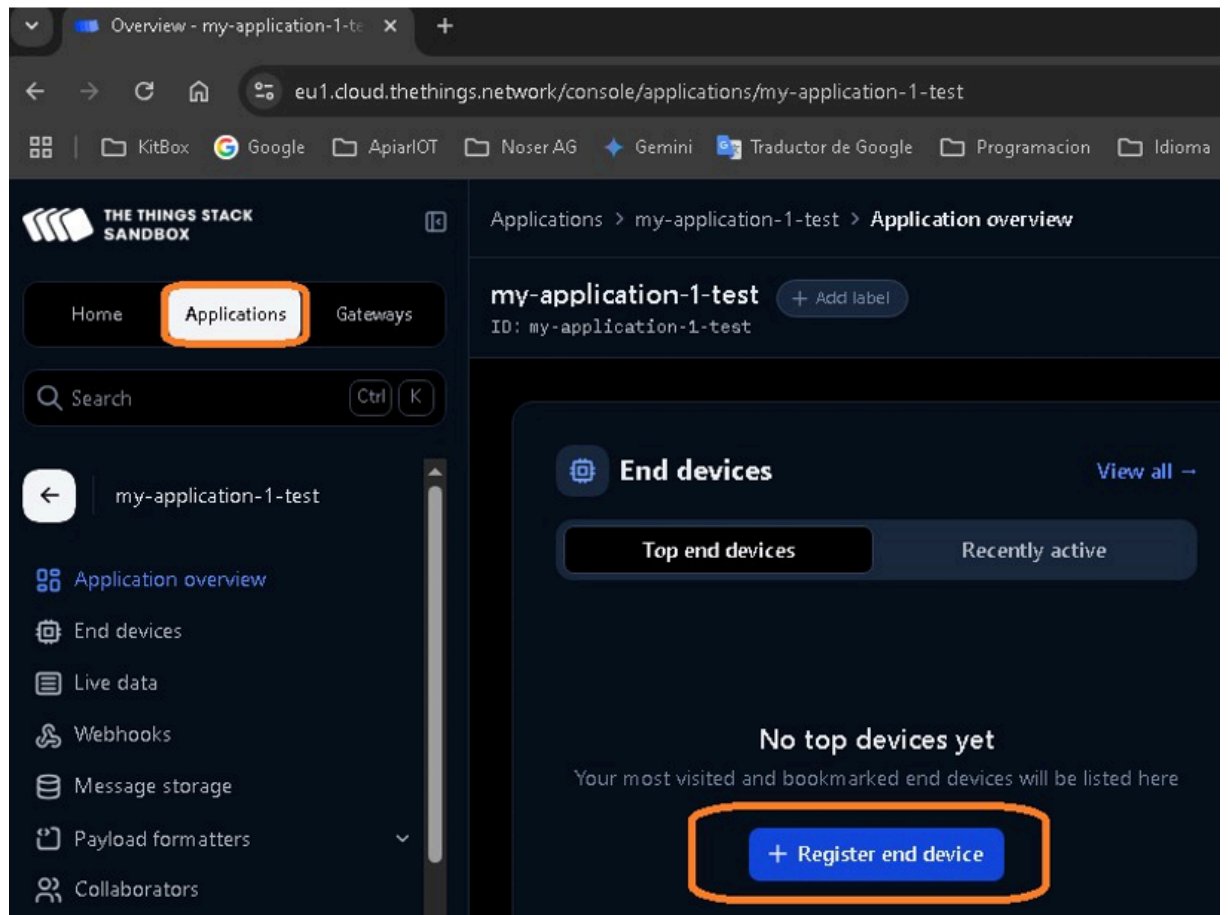
Select...



Add a label or [create one](#) to categorize your applications

Create application

- Registriere das "End-device".



- End device type, End device brand, Model, Hardware version, Firmware version, Profile region, Frequency plan

Register end device

Does your end device have a LoRaWAN® Device Identification QR Code? Scan it to speed up onboarding.

 Scan end device QR code

 [Device registration help](#)

End device type

Input method

☒ Select the end device in the LoRaWAN Device Repository

☐ Enter end device specifics manually

End device brand  *

STMicroelectronic...

Model  *

NUCLEO-WL55J...

Hardware Ver.  *

1.0

Firmware Ver.  *

LoRaWA...

Profile (Region) *

EU_863_870

NUCLEO-WL55JC1

LoRaWAN Specification 1.0.3, RP001 Regional Parameter (OTAA), Class A



The STMicroelectronics NUCLEO-WL55JC1 is a development board based on the STM32WL55JC microcontroller that comes with a built-in temperature sensor. The NUCLEO-WL55JC1 board supports a wide choice of Integrated Development Environments (IDE) and an expansion connector for Arduino UNO V3. Suitable for rapid prototyping by adding sensors and actuators.

[Product website](#) | [Data sheet](#)

LoRaWAN_End_v1.0.0

LoRaWAN_End_N v1.1.0

LoRaWAN_End_N v1.2.0

Other...

Frequency plan  *

Europe 863-870 MHz (SF9 for RX2 - recommended)

- Suche die Information von "LoRaWAN commissioning"

Provisioning information

JoinEUI  *

00.80.E1.15.05.00.D7.53

This end device can be registered on the network

DevEUI  *

11.11.11.11.11.11.11.11  0/50 used

AppKey  *

22.22.22.22.22.22.22.22.22.22.22.22.22.22.22.22 

End device ID  *

device-3-d87e

Device properties

Labels

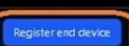
Select...

Add a label or [create one](#) to categorize your devices

After registration

☒ View registered end device

☐ Register another end device of this type



LoRaWAN Mode and Configuration

Reset Configuration

User Constants Platform Settings

LoRaWAN commissioning LoRaWAN middleware

LoRaWAN application

Configure the below parameters:

Search (Ctrl+F)

LoRaWAN device EUI 11.11.11.11.11.11.11.11

AppJoin EUI 00.80.E1.15.05.00.D7.53

Application key 22.22.22.22.22.22.22.22.22.22.22.22.22.22.22.22

Network key 00.00.00.00.00.00.00.00.00.00.00.00.00.00.00.00

Device Address 00.00.00.00

Network session key 00.80.E1.15.00.0A.D8.7E.00.80...

Application session key 00.80.E1.15.00.0A.D8.7E.00.80...

LoRaWAN application

Enable Key read access ☒

Current network ID

Applications > my-application-1-test > End devices > mdfsfdfsf > Device overview

mdfsdfsf ID: mdfsfdfsf + Add label No activity yet • ↑ n/a up / n/a down

Device overview Live data Messaging Location Payload formatters **Settings**

End device info Device repository --

NUCLEO-WL55JC1
STMicroelectronics International NV
Device website Data sheet
Light Pressure Temperature Humidity

Latest decoded payload See in live data --
No recent activity
There haven't been any uplinks recently in this application

device-3-d87e ID: device-3-d87e + Add label No activity yet • ↑ n/a up / n/a down

Network layer

LoRaWAN network-layer settings, behavior and session

Frequency plan ⓘ *
Europe 863-870 MHz (SF9 for RX2 - recommended) | ▾

LoRaWAN version ⓘ *
LoRaWAN Specification 1.0.1 | ▾

Regional Parameters version ⓘ *
TS001 Technical Specification 1.0.1 | ▾

LoRaWAN class capabilities ⓘ

☐ Supports class B

☐ Supports class C

Activation mode ⓘ *

☒ Over the air activation (OTAA)

☐ Activation by personalization (ABP)

☐ Define multicast group (ABP & Multicast)

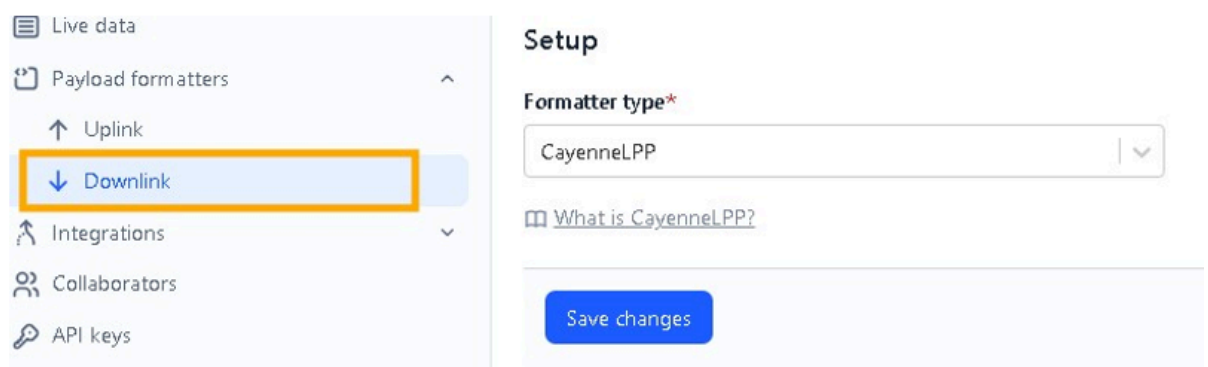
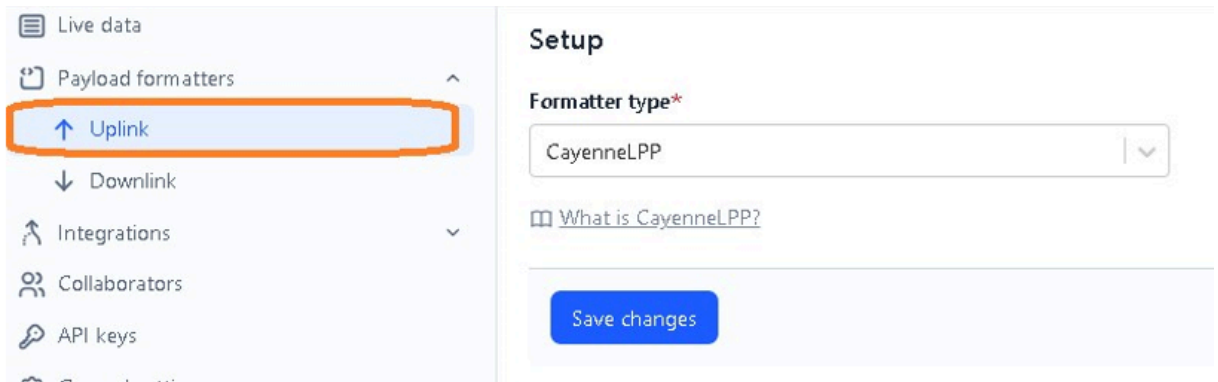
Session and MAC state reset ⓘ

Reset session and MAC state

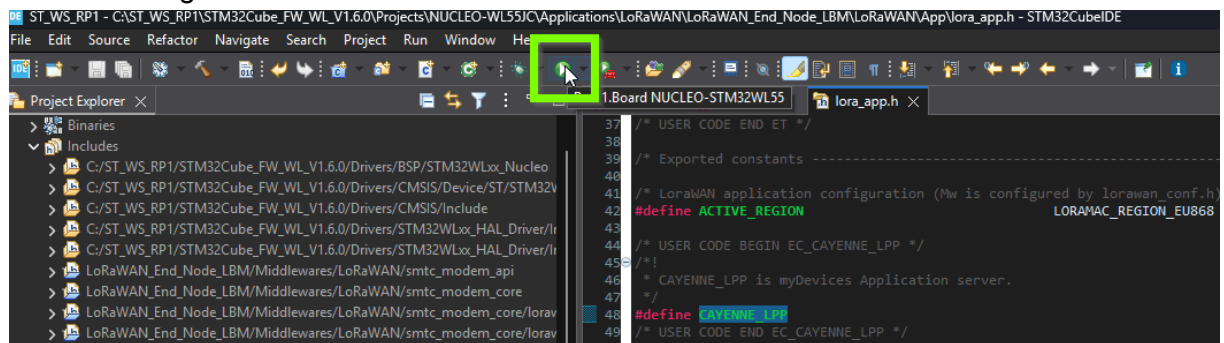
Advanced MAC settings ▾

Save changes

- Konfiguriere den "Uplink" und den "Downlink" für die "Payload" von "End-device"



- Lade das Programm auf das Board

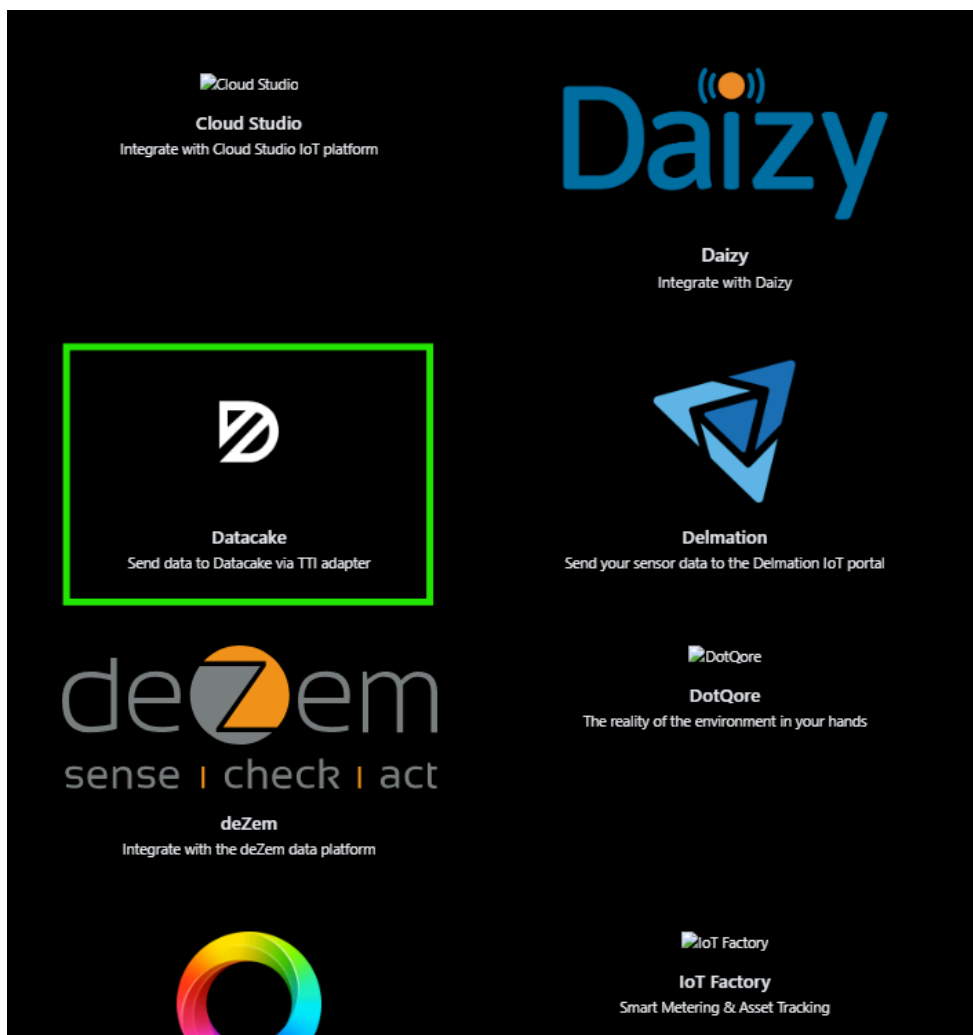
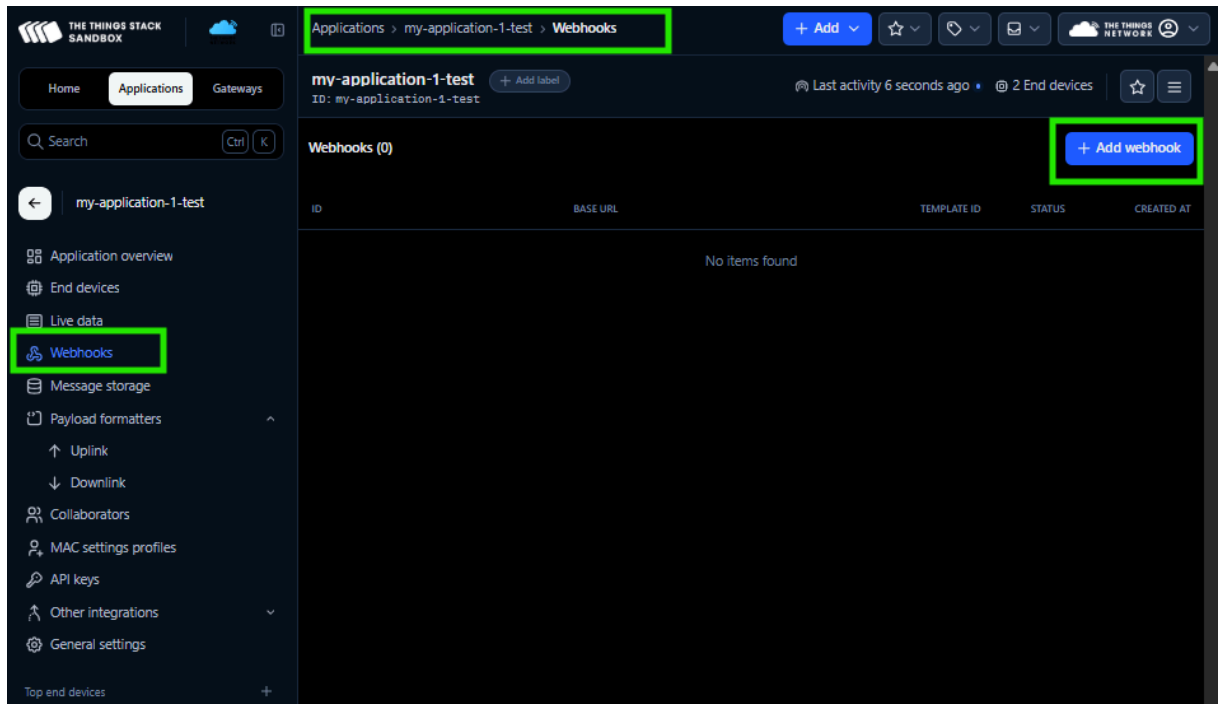


- Prüfe, ob TTN die Daten bekommt, wenn ja schalte dein Entwicklungsboard aus.

TIME	TYPE	DATA PREVIEW	Verbose stream	Export as JSON	Settings
13:37:52	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:52	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:41	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:41	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:33	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:33	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:26	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:26	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:16	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:16	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:09	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:09	Successfully processed data message	DevAddr: 26 08 FC AF			

Uplink forwarding TTN to Datacake

- Webhooks einrichten



The screenshot shows the 'my-application-1-test' page in the Things Stack Sandbox. The left sidebar contains a navigation menu with options like 'Application overview', 'End devices', 'Live data', 'Webhooks', 'Message storage', 'Payload formatters', 'Uplink', 'Downlink', 'Collaborators', 'MAC settings profiles', 'API keys', 'Other integrations', and 'General settings'. The main content area is titled 'my-application-1-test' and shows the 'Webhooks' section. A form titled 'Setup webhook for Datacake' is displayed, with the subtitle 'Send data to Datacake via TTI adapter' and a link 'About Datacake |'. The form has two input fields: 'Webhook ID' with the value 'my-new-datacake-webhook-1' and 'Token' which is highlighted with a red rectangle. Below the 'Token' field is the label 'Datacake API Token'. At the bottom of the form is a blue button labeled 'Create Datacake webhook'.

Applications > my-application-1-test > Webhooks > Add > Datacake

my-application-1-test [+ Add label](#)
ID: my-application-1-test

Search Ctrl K

my-application-1-test

- Application overview
- End devices
- Live data
- Webhooks
- Message storage
- Payload formatters
- Uplink
- Downlink
- Collaborators
- MAC settings profiles
- API keys
- Other integrations
- General settings

Setup webhook for Datacake
Send data to Datacake via TTI adapter
[About Datacake |](#)

Webhook ID *

my-new-datacake-webhook-1

Token *

Datacake API Token

Create Datacake webhook

- Du benötigst für den Token zuerst ein Konto bei Datacake.
- Erstelle dein Konto in [Datacake](#).

- Und füge einen "API User" hinzu

The screenshot shows the Datacake Administration interface. On the left is a sidebar with navigation options: Add Dashboard, Devices (All Devices, Manage Folders), Workspace (Gateways, Reports, Exports, Members, Rule Engine, Zones, Workspace, Integrations). The main content area is titled 'Members' and has tabs for 'In Workspace', 'Invited', and 'API Users'. The 'API Users' tab is active. In the top right corner, there is a blue button labeled 'Add API User' which is highlighted with a green rectangle. Below this, a table titled 'API Users' lists existing users.

API USER	API KEY	PERMISSIONS
usuario API Apiariot	***** 	
api-user-test-1	***** 	

Add API User



Name *

api-user-test-1

Workspace Permissions

☒ Select All

☐ Basics

Can manage basics like name and logo

☐ Cake Red

Can manage and view Cake Red

☐ Rules

Can manage rules

☐ Reports

Can manage reports

☐ Zones

Can manage zones

☐ Members

Can manage members

☒ Devices

Can add / remove devices

☐ Gateways

Can manage gateways

☐ Dashboards

Can manage dashboards

☐ Exports

Can manage exports

Device Permissions

All devices in workspace



☒ Can view the device and its data

☐ Can edit device metadata

☐ Can edit the device definition

☒ Can record measurements

Add individual device permission

Search

Device name

Filter by tags

Match at least one tag



Search tags...

Cancel

Add API User

- Kopiere das Token für TTN

The screenshot shows the Datacake Administration interface. On the left is a sidebar with navigation options like 'Add Dashboard', 'Devices', 'Workspace', 'Integrations', 'Organization', and 'Account'. The main area is titled 'Members' and has tabs for 'In Workspace', 'Invited', and 'API Users'. The 'API Users' tab is active, showing a table of API users. The table has columns for 'API USER', 'API KEY', and 'PERMISSIONS'. The user 'api-user-test-1' is listed, and its API key is masked with dots. A green box highlights the 'Copy' button next to the API key, and another green box highlights the 'Show' button next to it. The 'PERMISSIONS' column for this user shows 'Devices'.

API USER	API KEY	PERMISSIONS
U usuario API Apiarlot	***** Copy Show	Basics Members Devices +7
A api-user-test-1	***** Copy Show	Devices

- Du hast jetzt den Token für TTN

The screenshot shows the The Things Stack Sandbox interface. The left sidebar has tabs for 'Home', 'Applications', and 'Gateways'. The 'Applications' tab is active, showing a list of applications. The application 'my-application-1-test' is selected, and the 'Webhooks' section is active. The 'Setup webhook for Datacake' page is shown, with a form to create a new webhook. The 'Webhook ID' field is filled with 'my-new-datacake-webhook-1'. The 'Token' field is highlighted with a green box, and the text 'Datacake API Token' is visible below it. A 'Create Datacake webhook' button is at the bottom.

my-application-1-test
ID: my-application-1-test

Setup webhook for Datacake
Send data to Datacake via TTI adapter
[About Datacake](#)

Webhook ID *
my-new-datacake-webhook-1

Token *
Datacake API Token

Create Datacake webhook

- Jetzt bei Datacake, das Gerät hinzufügen

The screenshot shows the Datacake user interface. On the left is a sidebar with a user profile 'apiariot eemq@live.com' and various navigation links. The main area is the 'Devices' page under the 'Fleet' section. It includes a search bar, a 'Columns' dropdown, and a prominent blue '+ Add Device' button. Below these elements is a table with five columns: 'DEVICE', 'PRIMARY', 'SECONDARY', 'DEVICE SIGNAL', and 'DEVICE BATTERY'. The table is currently empty, with a message indicating that no devices are present in the workspace. At the bottom of the table area, it shows 'Showing 0 to 0 of 0 results' and a '50 per page' dropdown menu.

Add Device

You have 4 devices left in your plan. ✕

First, choose the connectivity type of your device.

- ☒  **LoRaWAN**
Choose from 15 LoRaWAN networks
- ☐  **API**
Generic API device with support for MQTT and HTTP connectivity
- ☐  **Pincode Claiming**
Claim an existing device by pincode
- ☐  **Dragino NB-IoT**
Connect Dragino NB-IoT devices
- ☐  **1NCE**
Connect 1NCE devices

Next

- Verwende als Netzwer Server: eu1.cloud.thethings.network (Schweiz)

Add LoRaWAN Device

You have 4 devices left in your plan. X

STEP 1Product

STEP 2Network Server

STEP 3Devices

STEP 4Plan

Datacake Product

You can add devices to an existing product on Datacake, create a new empty product or start with one of the templates. Products allow you to share the same configuration (fields, dashboard and more) between devices.

New Product from template

Create new product from a template

Existing Product

Add devices to an existing product

New Product

Create new empty product

Device Template

Datacake supports LoRaWAN devices from different manufacturers out of the box without complex configuration and setup.

caye

All Manufacturers

Cayenne LPP Starting Template

Datacake

Back

Next

SCHRITT 1
Produkt

SCHRITT 2
Netzwerk-Server

SCHRITT 3
Geräte

SCHRITT 4
Plan

Netzwerk-Server

Bitte wählen Sie den LoRaWAN Network Server, mit dem Ihre Geräte verbunden sind.

☐		Datacake LNS AUTOMATIC SETUP Einfach starten und skalieren mit einem verwalteten LNS	Uplinks	Downlinks
☒		The Things Stack V3 TTN V3 / Things Industries	Uplinks	Downlinks
☐		netmore ThingPark	Uplinks	Downlinks
☐		KPN	Uplinks	Downlinks
☐		LORIOT	Uplinks	Downlinks
☐		ChirpStack	Uplinks	Downlinks
☐		Helium Verwenden Sie Ihre eigene Konsole	Uplinks	Downlinks
Anzeigen von 1 bis 6 von 14 Ergebnissen			Vorherige	Weiter

Zurück

Weiter

You have 5 devices left in your plan. ✕

Add LoRaWAN Device

STEP 1
Product

STEP 2
Network Server

STEP 3
Devices

STEP 4
Plan

Add Devices

Manual

Import from The Things Stack

Please provide one or multiple LoRaWAN device EUIs along with the corresponding names they should have on Datacake.

Alternatively, you can choose to upload a CSV file that contains the DevEUI, device Name, location, and a set of tags. For more information on how to format the file, please refer to [our documentation](#).

📎 Drag and drop a .csv file here or click to choose one

DEVEUI

NAME

LOCATION

TAGS



00 80 E1 15 05 00 D7 53

8 bytes

Name1

Location

Add tag

Ac

+ Add another device

Back

Next

You have 5 devices left in your plan. ✕

Add LoRaWAN Device

STEP 1
Product

STEP 2
Network Server

STEP 3
Devices

STEP 4
Plan

Your organization plan (5 / 5 remaining)

Free

7 days data retention
500 datapoints / day

Back

Add 1 device

- Schalte deine "End Node" wieder ein.

The screenshot shows the Datacake web interface. On the left is a sidebar with navigation options: Add Dashboard, Manage Folders, Workspace (Gateways, Reports, Exports, Members, Rule Engine, Zones), and Organization. The main panel is titled 'Fleet > Devices' and shows a 'Devices' section with tabs for Overview, List, Grid, and Map. The 'List' tab is active, displaying a table with columns: DEVICE, PRIMARY, and SECONDARY. A single device is listed with ID '00.80.E1.15.05.00.D7.53', which is highlighted with a green box. The PRIMARY and SECONDARY columns show 'N/A'. Below the table, it says 'Showing 1 to 1 of 1 results'.

DEVICE	PRIMARY	SECONDARY
00.80.E1.15.05.00.D7.53	N/A	N/A

This screenshot shows the configuration page for a specific device in the Datacake interface. The browser address bar shows the URL: app.datacake.de/apiariot/devices/d/a7b3b80b-1c1b-42d8-ac99-d23ef0f5ab4b/t/dashboard-0. The page title is 'Fleet > 0080E115000AD87E'. The device ID '00.80.E1.15.05.00.D7.53' is displayed at the top. Below it, the 'Serial Number' is '0080E115000AD87E' and the 'Last update' is 'Sun Sep 27, 2026 21:38:05 GMT+02:00'. A navigation bar at the bottom of the main panel includes links for Dashboard, History, Downlinks, Configuration (highlighted with a green box), LoRaWAN, Debug, and Permissions. The left sidebar is identical to the first screenshot.

- Warte, bist Daten empfangen werden.

apiariot
eemq@live.com

+ Add Dashboard

Devices

All Devices

Manage Folders

Workspace

Gateways

Reports

Exports

Members

Rule Engine

Zones

Workspace

Integrations

Organization

Overview

Billing

Administrators

+ Add White Label Site

Account

Account Settings

Support

Documentation

Changelog

Basic device settings and metadata

Product & Hardware

Product settings and hardware-specific configuration

Fields

Configure measurement fields and data structure

Integrations

Third-party connections and data exports

Danger Zone

Device deletion and destructive actions

+ Add Field

Fields

Fields describe the data the device will store.

Live data

NAME	IDENTIFIER	TYPE	ROLE / SEMANTIC	CURRENT VALUE	LAST UPDATE
Temperature Sensor 0	TEMPERATURE_SENSOR_0	Numeric		0 °C	1 minute ago
Analog Input 0	ANALOG_INPUT_0	Numeric		0	1 minute ago
GPS 1	GPS_1	Location	Location	0.000000, 0.000000	1 minute ago

Suggested Fields

The device's decoder returned fields that have not yet been created on the product. You can review and create them from here.

Delete All Suggestions

IDENTIFIER	GUESSED TYPE	FIRST SEEN	LAST SEEN	LAST VALUE	
TEMPERATURE_SENSOR_1	Numeric	4 minutes ago	2 minutes ago	22	Create Field
HUMIDITY_2	Numeric	4 minutes ago	2 minutes ago	50	Create Field
BAROMETER_0	Numeric	4 minutes ago	2 minutes ago	1.000	Create Field

Configuration Fields

Configuration fields hold a static value and can have a product-wide default value, that can be overwritten on a device level. They can be accessed in decoders.

+ Add Configuration Field

NAME	IDENTIFIER	FIELD TYPE	DESCRIPTION	VALUE
------	------------	------------	-------------	-------

No fields have been created, yet

Create configuration fields to define configuration variables.

Add Field

Fields define the schema of the data the device stores.

Type

Numeric

Name

Temperature Sensor 1

Identifier

TEMPERATURE_SENSOR_1

The field identifier is a unique string that can consist of uppercase letters, numbers and underscores. Once a field has been created, the identifier can not be changed.

Unit

Optional

Role

None

Assign roles to highlight key measurement fields across the platform; each role is limited to one field per product.

Semantic

None

Assign semantic fields to standardize data across devices.

Formula

Optional

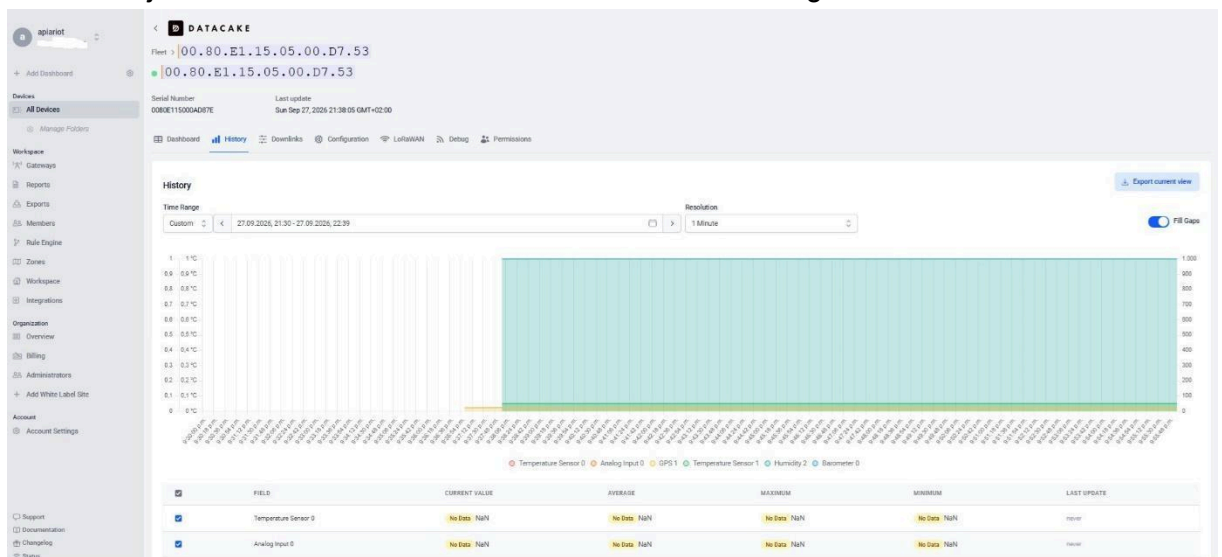
Formulas can be used to perform calculations on values from other fields. Fields that have a formula can not be written to from a decoder or via the API.

☐ Use Formula

Cancel

Add Field

- Du solltest jetzt in Datacake die Daten von deinem Entwicklungsboard sehen können.



- Und in TTN ebenfalls

The screenshot shows the TTN console interface for a device named 'device-3-d87e'. The 'Live data' tab is active, displaying a list of events. The interface includes a sidebar with navigation options like 'Application overview', 'End devices', 'Live data', 'Webhooks', 'Message storage', 'Payload formatters', 'Uplink', 'Downlink', 'Collaborators', 'MAC settings profiles', 'API keys', 'Other integrations', 'General settings', and 'Top end devices'. The main panel shows a table of events with columns for Time, Type, and Data Preview. The events include 'Schedule data downlink for transmission', 'Forward uplink data message', 'Successfully processed data message', 'Forward uplink data message', 'Forward join-accept message', 'Successfully processed join-request', and 'Accept join-request'. The 'Data Preview' column shows hexadecimal data and JSON payloads. A 'Console: Events cleared' message is visible at the bottom.

TIME	TYPE	DATA PREVIEW
15:39:06	Schedule data downlink for transmission	DevAddr: 26 08 C7 5C RxD Delay: 5
15:39:06	Forward uplink data message	DevAddr: 26 08 C7 5C Payload: { battery_voltage: 1, humidity: 1, light: 1, pressure: 1, temperature: 1 } FPort: 2 Data rate: SF8BW12S SNR: 12 RSSI:
15:39:06	Successfully processed data message	DevAddr: 26 08 C7 5C
15:39:06	Schedule data downlink for transmission	DevAddr: 26 08 C7 5C RxD Delay: 5
15:39:06	Forward uplink data message	DevAddr: 26 08 C7 5C Payload: { battery_voltage: 1, humidity: 1, light: 1, pressure: 1, temperature: 1 } FPort: 2 Data rate: SF12BW12S SNR: 9.8 RSSI:
15:39:06	Successfully processed data message	DevAddr: 26 08 C7 5C
15:39:40	Forward join-accept message	DevAddr: 26 08 C7 5C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63
15:39:39	Successfully processed join-request	DevAddr: 26 08 AD 2C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63
15:39:38	Accept join-request	DevAddr: 26 08 C7 5C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63
15:39:41	Console: Events cleared	The events list has been cleared